

Homework 4

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Problem 1 Recall Q2 of Homework 4: Let $p : S^3 \rightarrow S^2$ be the Hopf bundle and let $q : T^3 \rightarrow S^2$ be the quotient map collapsing the complement of a ball in the 3-dimensional torus $T^3 = S^1 \times S^1 \times S^1$ to a point. Show that $pq : T^3 \rightarrow S^2$ induces the zero map on homotopy and reduced homology, but is not homotopic to a constant map.

Reprove this as follows. The Hopf map $S^3 \rightarrow S^2$ becomes null-homotopic when continued to $\mathbb{CP}^\infty$ since it induces zero on H^2 . So the composition to $\mathbb{CP}^\infty$ is null-homotopic. Choose a null-homotopy, and lift that to the 2-stage Postnikov tower, to get a map from $T^3 \rightarrow K(\mathbb{Z}, 3)$. Show that the map is non-zero, and in fact the generator of $H^3(T^3; \mathbb{Z})$, independently of the null-homotopy chosen; whereas, obviously, a constant map to S^2 could be lifted to a constant map.

Note: Use the fact that the first k -invariant for S^2 is the generator of $H^4(K(\mathbb{Z}, 2); \mathbb{Z})$.

Solution: S^2 has only reduced homology in degree 2 so the only nonzero reduced homology map is $(pq)_* : \tilde{H}_2(T^3) \rightarrow \tilde{H}_2(S^2)$ but this map factors through $\tilde{H}^2(S^3)$ which is zero, so it is zero. Let $i : S^2 \rightarrow K(\mathbb{Z}, 2)$ classify the generator of $H^2(S^2)$, since $(pq)^*i = 0$ the composite $T^3 \xrightarrow{pq} S^2 \xrightarrow{i} K(\mathbb{Z}, 2)$ is null-homotopic.

Since $\pi_2(S^2) = \mathbb{Z}$ and $\pi_3(S^2) = \mathbb{Z}$, the third stage of the Postnikov tower for S^2 is a fibration $K(\mathbb{Z}, 3) \rightarrow P_3(S^2) \rightarrow K(\mathbb{Z}, 2)$ with k -invariant the generator of $H^4(K(\mathbb{Z}, 2); \mathbb{Z})$.

Choose a nullhomotopy of the map $T^3 \rightarrow K(\mathbb{Z}, 2)$. This gives a map $T^3 \times I \rightarrow K(\mathbb{Z}, 2)$ which restricts to $T^3 \times \{0\}$ as the nullhomotopy and to $T^3 \times \{1\}$ as the original map. Since $P_3(S^2)$ is a fibration over $K(\mathbb{Z}, 2)$ we can lift this homotopy to get a map $T^3 \times I \rightarrow P_3(S^2)$ which restricts to $T^3 \times \{0\}$ as some map $f : T^3 \rightarrow P_3(S^2)$ and to $T^3 \times \{1\}$ as a lift of the original map. Since the original map is nullhomotopic, the lift of the original map must factor through $K(\mathbb{Z}, 3)$ so we get a map $T^3 \rightarrow K(\mathbb{Z}, 3)$.

Such maps are classified by $[T^3, K(\mathbb{Z}, 3)] \cong H^3(T^3; \mathbb{Z}) \cong \mathbb{Z}$. Let $\alpha \in H^3(T^3; \mathbb{Z})$ be the class corresponding to the map $T^3 \rightarrow K(\mathbb{Z}, 3)$ we just constructed. The claim is that α is a generator. It suffices to show that the class $\beta \in H^3(S^3; \mathbb{Z})$ corresponding to the Hopf map is a generator, since α is the pullback of β along q and $q^* : H^3(S^3; \mathbb{Z}) \rightarrow H^3(T^3; \mathbb{Z})$ is an isomorphism.

The Hopf map $p : S^3 \rightarrow S^2$ represents the generator of $\pi_3(S^2) \cong \mathbb{Z}$. Since $P_3 S^2$ agrees with S^2 through homotopy degree 3, the induced map $S^2 \rightarrow P_3 S^2$ also identifies $\pi_3(S^2) \cong \pi_3(P_3 S^2)$. The long exact sequence in homotopy for the fibration $K(\mathbb{Z}, 3) \rightarrow P_3(S^2) \rightarrow K(\mathbb{Z}, 2)$ gives an isomorphism $\pi_3(P_3(S^2)) \cong \pi_3(K(\mathbb{Z}, 3)) \cong \mathbb{Z}$. Under this isomorphism, the class β corresponds to the generator of $\pi_3(K(\mathbb{Z}, 3))$, so β is a generator of $H^3(S^3; \mathbb{Z})$ as claimed.

Different choices of nullhomotopy of the map $T^3 \rightarrow K(\mathbb{Z}, 2)$ differ by a map

$$T^3 \rightarrow \Omega K(\mathbb{Z}, 2) \simeq K(\mathbb{Z}, 1)$$

by gluing the two nullhomotopies together along their common boundary. Such maps are classified by $[T^3, K(\mathbb{Z}, 1)] \cong H^1(T^3; \mathbb{Z})$. The glued map

$$X \times S^1 \rightarrow K(\mathbb{Z}, 2)$$

has associated cohomology class $a + bt \in H^2(X \times S^1; \mathbb{Z})$ where $a \in H^2(X; \mathbb{Z})$ is the class of the original map $X \rightarrow K(\mathbb{Z}, 2)$, $b \in H^1(X; \mathbb{Z})$ is the class of the change in nullhomotopy, and t is the generator of $H^1(S^1; \mathbb{Z})$. Since the first k -invariant for S^2 is $\pm u^2$ where u is the generator of $H^2(K(\mathbb{Z}, 2); \mathbb{Z})$, we compute

$$(a + bt)^2 = a^2 + 2abt + t^2 = 0$$

Thus the change in nullhomotopy does not change the resulting map $T^3 \rightarrow K(\mathbb{Z}, 3)$, so α is independent of the choice of nullhomotopy as claimed.

Problem 2 For a finitely generated Abelian group G and a simply connected space X , show that there is a short exact sequence

$$\text{Ext}^1(G; \pi_{n+1}X) \longrightarrow [M(G; n), X] \longrightarrow \text{Hom}(G; \pi_n X).$$

If $\pi_1 X \neq \{1\}$, one should also take its action on π_n into account.

Solution: Assume $n \geq 2$, so that the Moore space $M(G, n)$ is simply connected. Since G is finitely generated, choose a presentation

$$0 \rightarrow R \rightarrow F \rightarrow G \rightarrow 0$$

with F and R finitely generated free abelian groups. The Moore space $M(G, n)$ may be constructed as the cofiber of a map

$$\bigvee_R S^n \longrightarrow \bigvee_F S^n,$$

so it has cells only in dimensions n and $n + 1$, and

$$\tilde{H}_i(M(G, n); \mathbb{Z}) = \begin{cases} G, & i = n, \\ 0, & i \neq n. \end{cases}$$

Because $M(G, n)$ has dimension $n + 1$, any map $M(G, n) \rightarrow X$ only sees the $(n + 1)$ st Postnikov stage of X . Thus we may replace X by $P_{n+1}X$. Since X is simply connected, this Postnikov stage fits into a fibration

$$K(\pi_{n+1}X, n + 1) \longrightarrow P_{n+1}X \longrightarrow K(\pi_n X, n),$$

up to the lower Postnikov data, which is invisible to $M(G, n)$ because $M(G, n)$ is $(n - 1)$ -connected. Therefore a map $M(G, n) \rightarrow X$ has a primary invariant obtained by composing with

$$X \rightarrow K(\pi_n X, n).$$

This gives a natural map

$$[M(G, n), X] \longrightarrow H^n(M(G, n); \pi_n X).$$

By the universal coefficient theorem,

$$H^n(M(G, n); \pi_n X) \cong \text{Hom}(H_n(M(G, n)), \pi_n X) \cong \text{Hom}(G, \pi_n X),$$

since $H_{n-1}(M(G, n)) = 0$. This is the right-hand map in the desired sequence.

Now consider the kernel of this map. If the primary invariant is zero, then after choosing a nullhomotopy of the composite

$$M(G, n) \rightarrow X \rightarrow K(\pi_n X, n),$$

the map lifts to the homotopy fiber $K(\pi_{n+1} X, n + 1)$. Hence the kernel is identified with $H^{n+1}(M(G, n); \pi_{n+1} X)$. Applying the universal coefficient theorem again gives

$$0 \rightarrow \text{Ext}^1(H_n(M(G, n)), \pi_{n+1} X) \rightarrow H^{n+1}(M(G, n); \pi_{n+1} X) \rightarrow \text{Hom}(H_{n+1}(M(G, n)), \pi_{n+1} X) \rightarrow 0.$$

Since $H_n(M(G, n)) = G$ and $H_{n+1}(M(G, n)) = 0$, this reduces to

$$H^{n+1}(M(G, n); \pi_{n+1} X) \cong \text{Ext}^1(G, \pi_{n+1} X).$$

Therefore the kernel of $[M(G, n), X] \longrightarrow \text{Hom}(G, \pi_n X)$ is naturally $\text{Ext}^1(G, \pi_{n+1} X)$.

We obtain the short exact sequence

$$0 \rightarrow \text{Ext}^1(G, \pi_{n+1} X) \rightarrow [M(G, n), X] \rightarrow \text{Hom}(G, \pi_n X) \rightarrow 0.$$

Problem 3 The Bott periodicity theorem asserts that the stable, large n , values of $\pi_k U(n)$ are 0 and $\mathbb{Z}$, for k even/odd, with isomorphisms induced by the natural inclusions. Consider the fiber bundles

$$SU(n) \longrightarrow SU(n + 1) \longrightarrow S^{2n+1},$$

and recall that we showed in class that the projection to S^{2n+1} induces an isomorphism on integral H_{2n+1} .

1. Check that $\pi_k SU(n) \rightarrow \pi_k SU(n + 1)$ is an isomorphism for $k < 2n$ and onto for $k = 2n$, with cyclic kernel.
2. Use $\pi_4(S^3) = \mathbb{Z}/2$ to conclude that the generator of $H_5(SU(3); \mathbb{Z})$ is not spherical, not in the image of the Hurewicz map, and hence that $SU(3)$ is not homotopy equivalent to $S^3 \times S^5$, even though its homology could suggest that.

Solution: We use the fibration

$$SU(n) \longrightarrow SU(n+1) \longrightarrow S^{2n+1}.$$

The associated long exact sequence in homotopy contains

$$\pi_{k+1}(S^{2n+1}) \rightarrow \pi_k(SU(n)) \rightarrow \pi_k(SU(n+1)) \rightarrow \pi_k(S^{2n+1}).$$

If $k < 2n$, then $\pi_k(S^{2n+1}) = 0$ and $\pi_{k+1}(S^{2n+1}) = 0$ and so

$$\pi_k(SU(n)) \xrightarrow{\sim} \pi_k(SU(n+1))$$

is an isomorphism for $k < 2n$. If $k = 2n$, the relevant part of the long exact sequence is

$$\pi_{2n+1}(S^{2n+1}) \rightarrow \pi_{2n}(SU(n)) \rightarrow \pi_{2n}(SU(n+1)) \rightarrow \pi_{2n}(S^{2n+1}).$$

Since $\pi_{2n}(S^{2n+1}) = 0$ the map $\pi_{2n}(SU(n)) \rightarrow \pi_{2n}(SU(n+1))$ is surjective. Its kernel is the image of $\pi_{2n+1}(S^{2n+1}) \cong \mathbb{Z}$ and is therefore cyclic. This proves (1).

Now specialize to $n = 2$. We have the fibration

$$SU(2) \longrightarrow SU(3) \longrightarrow S^5.$$

Since $SU(2) \cong S^3$, this is

$$S^3 \longrightarrow SU(3) \longrightarrow S^5.$$

By the hypothesis recalled in the problem, the projection $SU(3) \rightarrow S^5$ induces an isomorphism

$$H_5(SU(3); \mathbb{Z}) \xrightarrow{\sim} H_5(S^5; \mathbb{Z}).$$

Suppose, for contradiction, that the generator of $H_5(SU(3); \mathbb{Z})$ is spherical. Then there is a map

$$f : S^5 \rightarrow SU(3)$$

such that $f_*[S^5]$ is a generator of $H_5(SU(3); \mathbb{Z})$. Composing with the projection gives a map

$$S^5 \xrightarrow{f} SU(3) \rightarrow S^5$$

of degree ± 1 . Therefore this composite is a homotopy equivalence of S^5 . After composing f with a homotopy inverse of this degree ± 1 map if necessary, we get a homotopy section of

$$SU(3) \rightarrow S^5.$$

But a homotopy section would force the connecting homomorphism

$$\partial : \pi_5(S^5) \rightarrow \pi_4(S^3)$$

in the long exact sequence of the fibration to be zero. Indeed, exactness gives $\partial \circ p_* = 0$, and if p has a section, then $p_* : \pi_5(SU(3)) \rightarrow \pi_5(S^5)$ is surjective, forcing $\partial = 0$.

We will show that ∂ is not zero. The relevant part of the long exact sequence is

$$\pi_5(SU(3)) \rightarrow \pi_5(S^5) \xrightarrow{\partial} \pi_4(S^3) \rightarrow \pi_4(SU(3))$$

By the first part, applied to $SU(3) \rightarrow SU(4)$, the group $\pi_4(SU(3))$ is already stable, because $4 < 6$. Bott periodicity gives $\pi_4(SU) = 0$ hence $\pi_4(SU(3)) = 0$. Therefore exactness implies that

$$\partial : \pi_5(S^5) \rightarrow \pi_4(S^3)$$

is surjective. Since $\pi_5(S^5) \cong \mathbb{Z}$ and $\pi_4(S^3) \cong \mathbb{Z}/2$, this connecting homomorphism is nonzero. Thus the fibration cannot have a homotopy section.

This contradicts the conclusion obtained from assuming that the generator of $H_5(SU(3); \mathbb{Z})$ was spherical. Hence the generator of $H_5(SU(3); \mathbb{Z})$ is not spherical, i.e. it is not in the image of the Hurewicz map

$$\pi_5(SU(3)) \rightarrow H_5(SU(3); \mathbb{Z}).$$

Finally, if $SU(3)$ were homotopy equivalent to $S^3 \times S^5$, then the generator of $H_5(SU(3); \mathbb{Z})$ would correspond to the fundamental class of the S^5 factor, hence would be spherical. This contradicts what we just proved. Therefore

$$SU(3) \not\cong S^3 \times S^5.$$